

```
> sudo nano /usr/local/bin/system_info
```

```
Password:
```

```
~/Doc/AU/FALL_2026/Information_Security/home
```

```
29s
```

```
> sudo chmod +x /usr/local/bin/system_info
```

```
~/Doc/AUCA/FALL_2026/Information_Security/home
```

```
> system_info
```

```
Hostname: esq1res-MacBook-Pro.local
```

```
Current User: arsen
```

Disk Space:	Filesystem	Size	Used	Avail	Capacity
ty iused ifree %iused	Mounted on				
/dev/disk3s1s1	926Gi	12Gi	635Gi	2%	459k 4.
3G 0%	/				
devfs	200Ki	200Ki	0Bi	100%	691
0 100%	/dev				
/dev/disk3s6	926Gi	1.0Gi	635Gi	1%	1 6.
7G 0%	/System/Volumes/VM				
/dev/disk3s2	926Gi	8.8Gi	635Gi	2%	2.8k 6.
7G 0%	/System/Volumes/Preboot				
/dev/disk3s4	926Gi	7.5Mi	635Gi	1%	110 6.
7G 0%	/System/Volumes/Update				
/dev/disk1s2	500Mi	6.0Mi	481Mi	2%	1 4.
9M 0%	/System/Volumes/xarts				
/dev/disk1s1	500Mi	5.6Mi	481Mi	2%	32 4.
9M 0%	/System/Volumes/iSCPreboot				
/dev/disk1s3	500Mi	2.9Mi	481Mi	1%	84 4.
9M 0%	/System/Volumes/Hardware				
/dev/disk3s5	926Gi	269Gi	635Gi	30%	3.3M 6.
7G 0%	/System/Volumes/Data				
map auto_home	0Bi	0Bi	0Bi	100%	0
0 -	/System/Volumes/Data/home				

1. Create a script to display a greeting

```
> nano greet.sh
```

```
~/Doc/AU/FALL_2026/Information_Security/home
```

18s

```
> ls
```

```
greet.sh
```

```
~/Doc/AUCA/FALL_2026/Information_Security/home
```

```
> |
```

UW PICO 5.09

File: greet.sh

```
#!/bin/bash
```

```
echo "Hello, $(whoami)!"
```

```
~/Doc/AU/FALL_2026/Information_Security/home
```

13s

```
> chmod +x greet.sh
```

```
~/Doc/AUCA/FALL_2026/Information_Security/home
```

```
> ./greet.sh
```

```
Hello, arsen!
```

```
~/Doc/AUCA/FALL_2026/Information_Security/home
```

```
> |
```

2. Create a script that displays time remaining before work day ends

```
> nano current_time.sh
```

```
~/Doc/AU/FALL_2026/Information_Security/home
```

```
> ls
```

```
current_time.sh
```

```
~/Doc/AUCA/FALL_2026/Information_Security/home
```

```
> |
```

UW PICO 5.09

File: current_time.sh

Modified

```
#!/bin/bash
```

```
current_time=$(date +%H:%M)
```

```
current_hour=$(date +%H)
```

```
current_minute=$(date +%M)
```

```
remaining_minutes=$((18 * 60 - 10#$current_hour * 60 - 1$
```

```
hours=$((remaining_minutes / 60))
```

```
minutes=$((remaining_minutes % 60))
```

```
echo "Current time: $current_time"
```

```
$ minutes."
```

```
> ./current_time.sh
```

```
Current time: 13:36
```

```
Work day ends after 4 hours and 24 minutes.
```

3. Create a script to count word occurrences in a file

```
> nano count_word.sh
```

```
~/Doc/AU/FALL_2026/Information_Security/home 10s
```

```
> chmod +x count_word.sh
```

```
~/Doc/AUCA/FALL_2026/Information_Security/home
```

```
> |
```

UW PICO 5.09

File: count_word.sh

Modified

```
#!/bin/bash
```

```
file=$1
```

```
word=$2
```

```
count=$(grep -o -w "$word" "$file" | wc -l)
```

```
echo "The word '$word' appears $count times."
```

```
~/Doc/AUCA/FALL_2026/Information_Security/home
```

```
> echo "apple banana apple orange apple" > test.txt
```

```
~/Doc/AUCA/FALL_2026/Information_Security/home
```

```
> ./count_word.sh test.txt apple
```

```
The word 'apple' appears      3 times.
```

```
~/Doc/AUCA/FALL_2026/Information_Security/home
```

```
> |
```

4. Create a Script to find and delete empty files

```
> nano delete_empty_files.sh
```

```
~/Doc/AUCA/FALL_2026/Information_Security/home
```

```
> chmod +x delete_empty_files.sh
```

```
UW PICO 5.09 File: delete_empty_files.sh Modified
```

```
#!/bin/bash
```

```
directory=$1
```

```
find "$directory" -type f -empty -print -delete
```

```
~/Doc/AUCA/FALL_2026/Information_Security/home
> touch test_folder/empty1.txt

~/Doc/AUCA/FALL_2026/Information_Security/home
> touch test_folder/empty2.txt

~/Doc/AUCA/FALL_2026/Information_Security/home
> echo "Hello" > test_folder/file.txt

~/Doc/AUCA/FALL_2026/Information_Security/home
> ls -l test_folder
total 8
-rw-r--r--@ 1 arsen  staff  0 Sep 15 13:40 empty1.txt
-rw-r--r--@ 1 arsen  staff  0 Sep 15 13:40 empty2.txt
-rw-r--r--@ 1 arsen  staff  6 Sep 15 13:40 file.txt

~/Doc/AUCA/FALL_2026/Information_Security/home
> ./delete_empty_files.sh test_folder
test_folder/empty2.txt
test_folder/empty1.txt

~/Doc/AUCA/FALL_2026/Information_Security/home
> ls -l test_folder
total 8
-rw-r--r--@ 1 arsen  staff  6 Sep 15 13:40 file.txt
```